

annotated
version

MACHINE LEARNING COURSE · CS-433

Least Squares

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Martin Jaggi

credits to Mohammad Emtiyaz Khan & Rüdiger Urbanke

EPFL

Motivation

In rare cases, one can compute the optimum of the cost function analytically. Linear regression using a mean-squared error cost function is one such case. Here the solution can be obtained explicitly, by solving a linear system of equations. These equations are sometimes called the normal equations. This method is one of the most popular methods for data fitting. It is called least squares.

How we will get there

Plan ① show $\mathcal{L}(w)$ convex
② $\nabla \mathcal{L}(w) \stackrel{!}{=} 0$

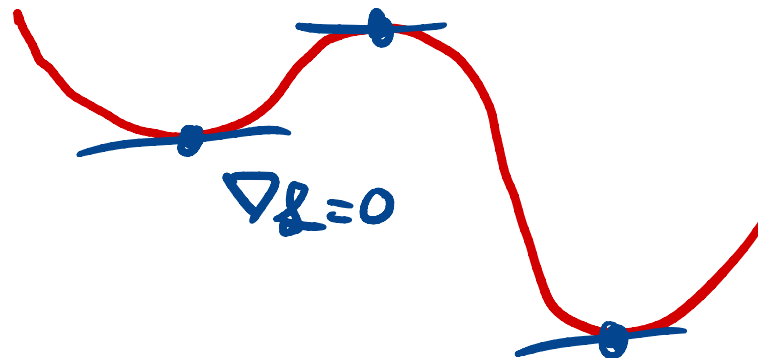
To derive the normal equations, we first show that the problem is convex. We then use the optimality conditions for convex functions (see the previous lecture on optimization). I.e., at the optimum parameter, call it $w^\star$, it must be true that the gradient of the cost function is 0:

$\Rightarrow w$ is optimal

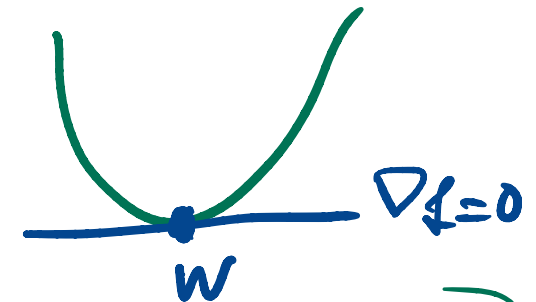
$$\nabla \mathcal{L}(w^\star) = 0$$

This is a system of D equations.

non-convex



convex



Normal equations

Recall that the cost function for linear regression with mean-squared error is given by

$$\mathcal{L}(\mathbf{w}) = \frac{1}{2N} \sum_{n=1}^N \left(y_n - \overset{\text{f}_w(\mathbf{x})}{\overset{\parallel}{\mathbf{x}_n^\top \mathbf{w}}} \right)^2 = \frac{1}{2N} \overset{e}{\underbrace{(\mathbf{y} - \mathbf{X}\mathbf{w})}^\top} \overset{e}{\underbrace{(\mathbf{y} - \mathbf{X}\mathbf{w})}}$$

$= \frac{1}{2N} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix},$$

$$\mathbf{X} = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1D} \\ X_{21} & X_{22} & \dots & X_{2D} \\ \vdots & \vdots & \ddots & \vdots \\ X_{N1} & X_{N2} & \dots & X_{ND} \end{bmatrix}$$

loss
feature $\in \mathbb{R}^N$

← row of $\mathbf{x}_2$ datapoint $\in \mathbb{R}^D$

The cost is convex in w

There are several ways of proving this.

1. Simplest way: observe that L is naturally represented as the sum (with positive coefficients) of the simple terms $(y_n - x_n^\top w)^2$. Each of these is the composition of a linear function with a convex function (the square function). Therefore, each term is convex and hence the sum is convex.

$$\sum_n$$

$\Rightarrow L(w)$ convex

$$f \circ g = f(g(w))$$

$\cdot g: \text{linear}$

$\cdot f: \text{convex}$

$\Rightarrow f \circ g$ convex

Rule

The cost is convex in w

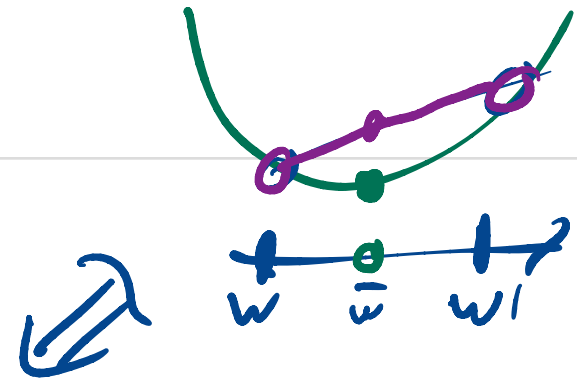
2. Directly verify the definition, that for any $\lambda \in [0, 1]$ and w, w' ,

$$\underbrace{\mathcal{L}(\lambda w + (1 - \lambda)w')}_{\text{L}} - \underbrace{(\lambda \mathcal{L}(w) + (1 - \lambda)\mathcal{L}(w'))}_{\text{line segment}} \leq 0$$

Computation: LHS =

$$= -\frac{1}{2N} \lambda(1 - \lambda) \underbrace{\|X(w - w')\|_2^2}_{\|\dots\|^2}$$

which indeed is non-positive.



$$\begin{aligned} & \|X\bar{w} - y\|^2 \\ & - 2\lambda\|Xw - y\|^2 \\ & - (1-\lambda)\|Xw' - y\|^2 \end{aligned}$$

.....

The cost is convex in w

3. We can compute the second derivative (the Hessian) and show that it is positive semidefinite (all its eigenvalues are non-negative). For the present case a computation shows that the Hessian has the form

$$\begin{aligned}\nabla \mathcal{L} &= -\frac{1}{N} X^T (y - Xw) \\ \nabla^2 \mathcal{L} &= \frac{1}{N} X^T X \quad \text{is}\end{aligned}$$

This matrix is indeed positive semidefinite since its non-zero eigenvalues are the squares of the non-zero singular values of the matrix X .

$$\begin{aligned}w^T M w &= w^T X^T X w \\ &= \|Xw\|^2 \geq 0 \quad \checkmark\end{aligned}$$

$$\begin{aligned}\nabla^2 \mathcal{L} &= \left(\frac{\partial^2 \mathcal{L}}{\partial w_i \partial w_j} \right)_{ij} \\ &\succeq 0 \quad \text{p.s.d.} \\ &\Rightarrow \mathcal{L} \text{ convex} \quad \left(w^T M w \geq 0 \quad \forall w \right)\end{aligned}$$

not ok

step ②

The normal equations

$$\nabla \mathcal{L}(\mathbf{w}) \stackrel{!}{=} \mathbf{0} \in \mathbb{R}^D$$

Now that we know the function is convex, let us find its minimum. If we take the gradient with respect to the weight vector $\mathbf{w}$ we get

$$\nabla \mathcal{L}(\mathbf{w}) = -\frac{1}{N} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{w}) \stackrel{!}{=} \mathbf{0}$$

If we set this expression to 0 we get the **normal equations for linear regression**,

$$\Leftrightarrow \mathbf{X}^\top \underbrace{(\mathbf{y} - \mathbf{X}\mathbf{w})}_{\substack{\text{error} \\ \in \mathbb{R}^N}} = \mathbf{0}$$

$$\Leftrightarrow \mathbf{X}^\top \mathbf{X} \mathbf{w} = \mathbf{X}^\top \mathbf{y}$$
$$\mathbf{A} \mathbf{w} = \mathbf{b}$$

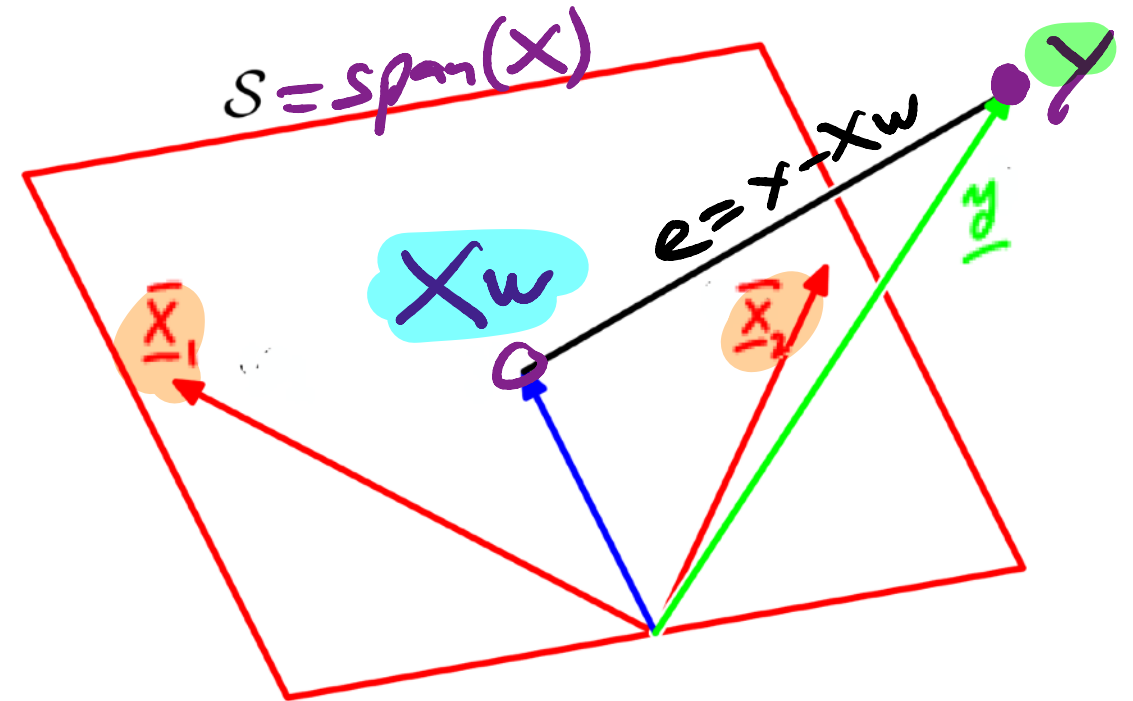
D equations
D variables
linear system

Geometric interpretation

The $\underline{e}$ is orthogonal to all $\text{all rows of } X^T$ columns of X .

The **span** of X is the space spanned by its columns; every element can be written as $u = Xw$.

The normal equations tell us that the optimum choice $u^\star$ is the element for which $y - u^\star$ is orthogonal to $\text{span}(X)$. In other words, $u^\star$ is the projection of y onto $\text{span}(X)$.



Taken from Bishop's book.

$$\text{span}(X) = \{Xw \mid w \in \mathbb{R}^D\}$$

$$\min_w \mathcal{L}(w) = \frac{1}{2} e^T e = \frac{1}{2} \|y - Xw\|^2$$

Least squares

The matrix $X^T X \in \mathbb{R}^{D \times D}$ is called the **Gram matrix**. If it is invertible, we can multiply the normal equation by its inverse from the left to get a closed-form expression for the minimum:

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

We can use this model to predict a new value for an unseen datapoint (test point) $\mathbf{x}_m$:

$$\hat{y}_m := \mathbf{x}_m^T \mathbf{w}^* = \mathbf{x}_m^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$f_{\mathbf{w}}(\mathbf{x}_m)$$

Invertibility and uniqueness

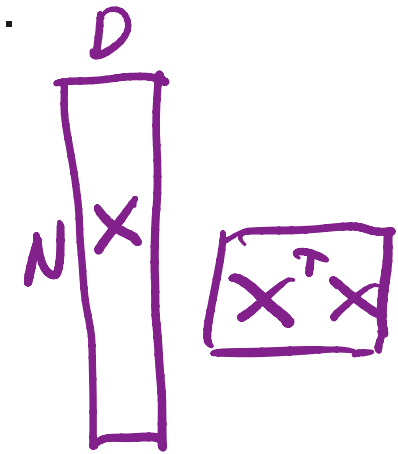
CASE
 $D \ll N$:

lemma

The Gram matrix $X^T X \in \mathbb{R}^{D \times D}$ is invertible if and only if X has full column rank, i.e. $\text{rank}(X) = D$.



Proof: Assume first that $\text{rank}(X) < D$. Then there exists a non-zero vector u with $Xu = 0$. It follows that $X^T Xu = 0$, and so $\text{rank}(X^T X) < D$. Therefore $X^T X$ is not invertible.



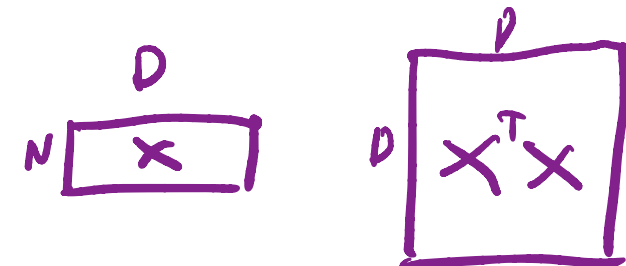
Conversely, assume $X^T X$ is not invertible. Then there is a non-zero v with $X^T Xv = 0$, and

CASE
 $D \gg N$:

$$0 = v^T X^T X v = (Xv)^T (Xv) = \|Xv\|^2$$

This implies $Xv = 0$, i.e. $\text{rank}(X) < D$.

over-parameterized



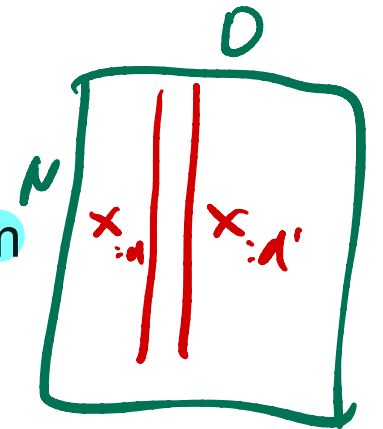
Rank deficiency and ill-conditioning

Unfortunately, in practice, X is often **rank deficient**.

over-parameterized

- If $D > N$, we always have $\text{rank}(X) < D$ (since row rank = column rank)
- If $D \leq N$, but some of the columns $x_{:d}$ are (nearly) collinear, then the matrix is **ill-conditioned**, leading to numerical issues when solving the linear system.

Can we solve least squares if X is rank deficient? Yes, using a **linear system solver**.



*$\Rightarrow$ don't invert $X^T X$
but use L.S.S.*

$$X^T X w = X^T y$$

Summary of linear regression

We have studied three types of methods:

1. Grid search

2. Iterative optimization algorithms

(stochastic) gradient descent

3. Least squares

closed-form solution, for linear MSE

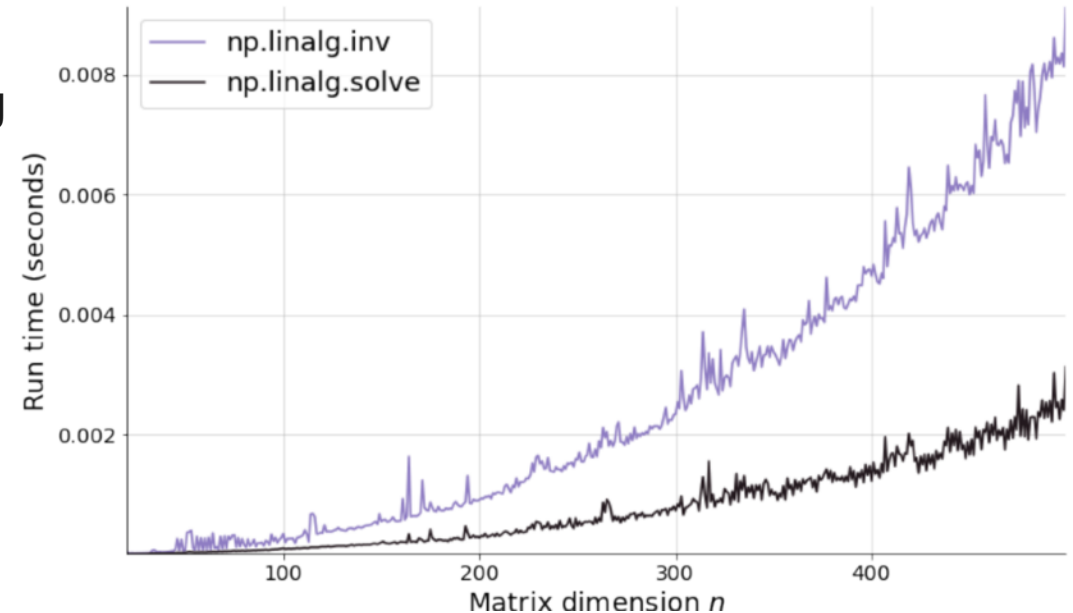
today

Solving linear systems

There are many ways to solve a linear system $Aw = b$, but it usually involves a decomposition of the matrix A such as the QR or LU decomposition, which are very robust. Matlab's backslash operator and NumPy's linalg package implement this in one line:

```
w = np.linalg.solve(A, b)
```

It is important to never invert a matrix to solve a linear system — this would incur a cost at least three times that of using a linear solver.



For a robust implementation, see Sec. 7.5.2 of Kevin Murphy's book.

Closed-form solution for MAE

Can you derive a closed-form solution for the 1-parameter model when using the MAE cost function?

See this short article: johnmyleswhite.com/notebook/2013/03/22/modes-medians-and-means-an-unifying-perspective